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41 ively (Left) skewed\n")
42 -
43 tric\n")
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48 ----
49 me:", min(response_time), "ms\n")
50 : ", max(response_time), "ms\n")
51 re between 250 ms and 330 ms.\n")
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```
R 4.2.3 ~/\n> cat("Maximum Response Time: 380 ms\n")
Maximum Response Time: 380 ms
> cat("Most response times are between 250 ms and 330 ms.\n")
Most response times are between 250 ms and 330 ms.
>
> # -----
> # 5. Better Representation
> # -----
> cat("\nBetter Plot:\n")

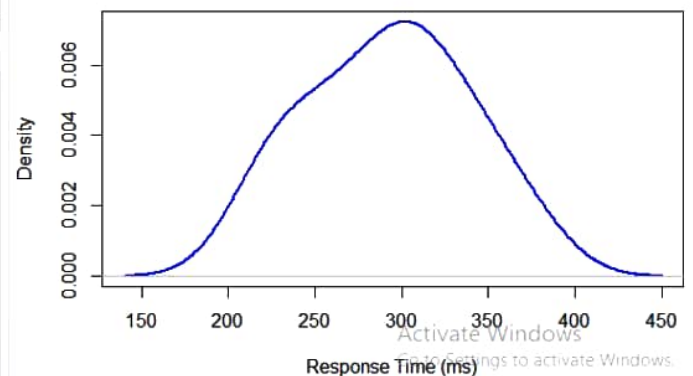
Better Plot:
> cat("Density Plot better represents the distribution because it provides a smooth view of the data pattern.\n")
Density Plot better represents the distribution because it provides a smooth view of the data pattern.
> |
```

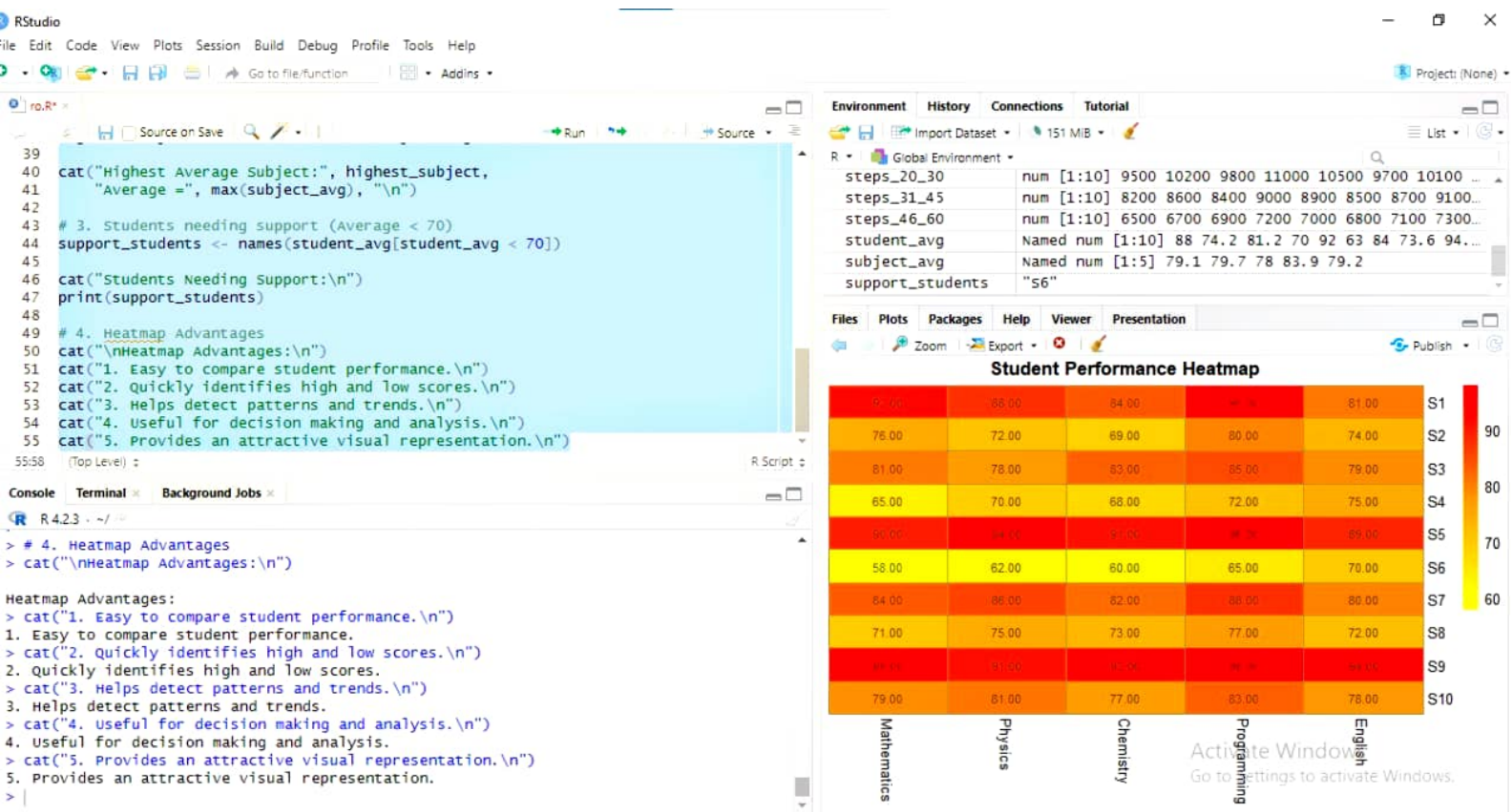
Environment	History	Connections	Tutorial
Global Environment	Import Dataset	151 MiB	
response_time	num [1:20]	210 225 230 240 250 260 270 280 285 290...	
steps_20_30	num [1:10]	9500 10200 9800 11000 10500 9700 10100 ...	
steps_31_45	num [1:10]	8200 8600 8400 9000 8900 8500 8700 9100...	
steps_46_60	num [1:10]	6500 6700 6900 7200 7000 6800 7100 7300...	
student_avg	Named num [1:10]	88 74.2 81.2 70 92 63 84 73.6 94...	
subject_avg	Named num [1:5]	79.1 79.7 78 83.9 79.2	

Files Plots Packages Help Viewer Presentation

Zoom Export Publish

Density Plot of Website Response Time





File Edit Code View Plots Session Build Debug Profile Tools Help

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61 # 4. Highest Median Group
62 # -----
63 medians <- c(
64   median(steps_20_30),
65   median(steps_31_45),
66   median(steps_46_60)
67 )
68 groups <- c("20-30", "31-45", "46-60")
69
70 cat("Median values:\n")
71 print(medians)
72
73
74 cat("\nHighest Median Group:",
75     groups[which.max(medians)], "\n")
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23 legend=colnames(electricity),
24 col=1:5,
25 lty=1,
26 pch=16)
27
28 # Average Consumption
29 avg <- colMeans(electricity)
30 print(avg)
31 cat("Month with Highest Average Consumption:", names(which.max(avg)), "\n")
32
33 # Standard Deviation
34 spread <- apply(electricity,2,sd)
35 print(spread)
36 cat("Month with Widest Spread:", names(which.max(spread)), "\n")
36:65 (Top Level)

Console Terminal Background Jobs
R 4.2.3 ~ /
> legend=colnames(electricity),
+ col=1:5,
+ lty=1,
+ pch=16)
>
> # Average Consumption
> avg <- colMeans(electricity)
> print(avg)
Jan Feb Mar Apr May
230.8 239.9 256.5 281.2 300.7
> cat("Month with Highest Average Consumption:", names(which.max(avg)), "\n")
Month with Highest Average Consumption: May
>
> # Standard Deviation
> spread <- apply(electricity,2,sd)
> print(spread)
Jan Feb Mar Apr May
9.223159 6.279597 5.380004 5.865151 5.538752
> cat("Month with Widest Spread:", names(which.max(spread)), "\n")
Month with Widest Spread: Jan
>
```

Environment History Connections Tutorial

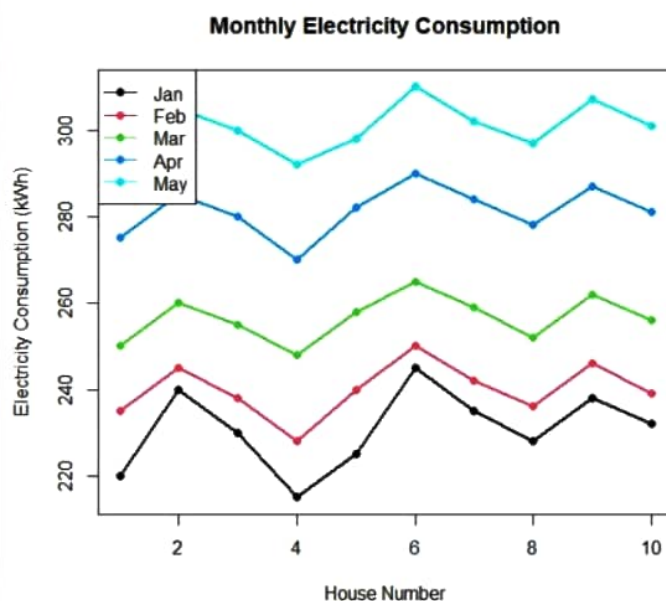
Import Dataset 178 M.B

R Global Environment

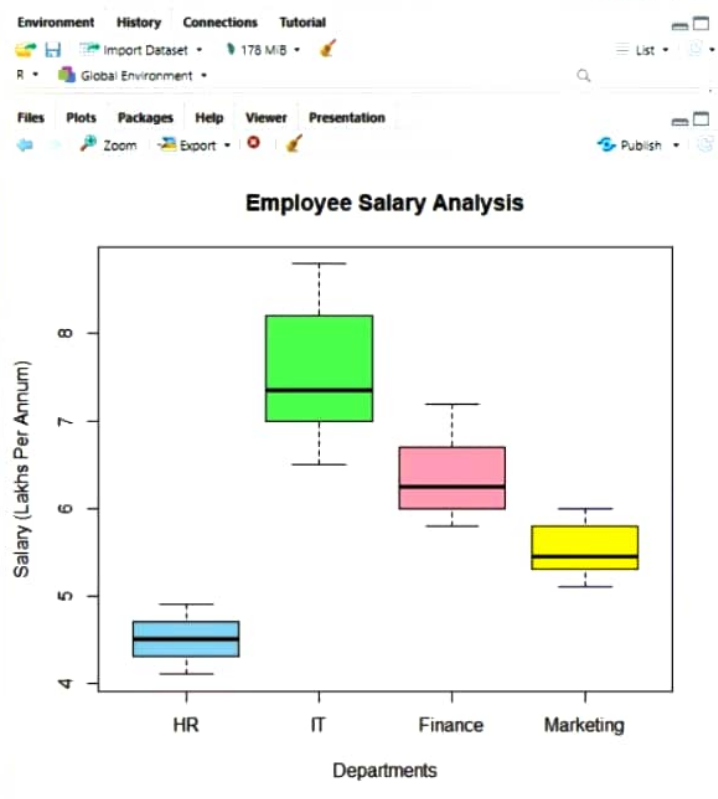
Files Plots Packages Help Viewer Presentation

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RStudio
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12 col=c("lightblue", "lightgreen", "pink", "yellow"),
13 main="Employee Salary Analysis",
14 xlab="Departments",
15 ylab="Salary (Lakhs Per Annum)")
16
17 # Standard Deviation
18 variation <- apply(salary, 2, sd)
19 print(variation)
20 cat("Department with Largest Variation:", names(which.max(variation)), "\n")
21
22 # Median
23 med <- apply(salary, 2, median)
24 print(med)
25 cat("Department with Highest Median Salary:", names(which.max(med)), "\n")
25:75 (Top Level)
R Script
Console Terminal Background Jobs
R 4.2.3 ~ /
> col=c("lightblue", "lightgreen", "pink", "yellow"),
+ main="Employee Salary Analysis",
+ xlab="Departments",
+ ylab="Salary (Lakhs Per Annum)")
>
> # Standard Deviation
> variation <- apply(salary, 2, sd)
> print(variation)
      HR      IT  Finance Marketing
0.2581989 0.7554248 0.4552167 0.3011091
> cat("Department with Largest Variation:", names(which.max(variation)), "\n")
Department with Largest Variation: IT
>
> # Median
> med <- apply(salary, 2, median)
> print(med)
      HR      IT  Finance Marketing
4.50    7.35    6.25    5.45
> cat("Department with Highest Median Salary:", names(which.max(med)), "\n")
Department with Highest Median Salary: IT
>
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RStudio
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1 # Monthly Income
2 income <- c(18000,22000,24000,26000,28000,
3            30000,34000,42000,85000,200000)
4
5 # Histogram
6 hist(income,
7      main = "Histogram of Monthly Income",
8      xlab = "Income",
9      col = "lightgreen")
10
11 # Q-Q Plot
12 qqnorm(income,
13        main = "Normal Q-Q Plot of Income")
14 qqline(income, col = "red", lwd = 2)
```

14:37 (Top Level) R Script

```
R 4.2.3 ~ ./
> qqnorm(waiting,
+        main = "Normal Q-Q Plot of waiting time")
> qqline(waiting, col = "red", lwd = 2)
> # Monthly Income
> income <- c(18000,22000,24000,26000,28000,
+            30000,34000,42000,85000,200000)
>
> # Histogram
> hist(income,
+      main = "Histogram of Monthly Income",
+      xlab = "Income",
+      col = "lightgreen")
>
> # Q-Q Plot
> qqnorm(income,
+        main = "Normal Q-Q Plot of Income")
> qqline(income, col = "red", lwd = 2)
>
```

